# IoT enabled adaptive clustering based energy efficient routing protocol for wireless sensor networks


## Muhammad Asad*

School of Computer Science and Technology,
Dalian University of Technology,
Dalian, 116000, China
Email: asadi244@gmail.com
*Corresponding author

## Muhammad Aslam

Department of Computer Science,
COMSATS University Islamabad,
WAH CANTT, 47040, Pakistan
Email: aslamhayat@mail.dlut.edu.cn

## Yao Nianmin and Naeem Ayoub

School of Computer Science and Technology,
Dalian University of Technology,
Dalian, 116000, China
Email: lucos@dlut.edu.cn
Email: nomi@mail.dlut.edu.cn

## Khalid Ibrahim Qureshi

School of Software Engineering,
Dalian University of Technology,
Dalian, 116000, China
Email: khalidibrahim@mail.dlut.edu.cn

## Ehsan Ullah Munir

Department of Computer Science,
COMSATS University Islamabad,
WAH CANTT, 47040, Pakistan
Email: ehsanmunir@comsats.edu.pk


**Abstract:** In this paper, we propose a new internet of things (IoT) enabled multi adaptive clustering (MAC) energy efficient routing protocol for wireless sensor networks (WSNs) to minimise energy dissipation and improve the network performance. This new technique holds the hybrid cluster formation algorithm in which the network topology is divided in two regions where the first region is centralised and the second region is distributed. Both regions contain homogeneous and heterogeneous nodes while the sink is static and located in the centre of both networks. Specifically, proposed IoT enabled MAC routing protocol holds the major three properties: enabling of resources to sensor nodes through IoT, hybrid cluster formation to distribute the network load evenly among sensor nodes and a new mechanism to minimise the energy consumption in long range data transmission. In all of our experiments MAC achieves greater performance than state-of-the-art routing protocols. Moreover, performance evaluation proofs that MAC outperforms in critical network environment.

**Keywords:** internet of things; IoT; wsnS; wireless sensor networks; energy efficient; routing protocols.

**Reference** to this paper should be made as follows: Asad, M., Aslam, M., Nianmin, Y., Ayoub, N., Qureshi, K.I. and Munir, E.U. (2019) 'IoT enabled adaptive clustering based energy efficient routing protocol for wireless sensor networks', *Int. J. Ad Hoc and Ubiquitous Computing*, Vol. 32, No. 2, pp.133–145.

Copyright © 2019 Inderscience Enterprises Ltd.

**Biographical notes:** Muhammad Asad received his Bachelor degree in Telecommunication and Networks from COMSATS Institute of Information Technology, WAH CANTT, Pakistan in 2014. Now he is pursuing his Masters in School of Computer Science and Technology, Dalian University of Technology, China under the supervision of Prof. Yao Nianmin. His main research interests include wireless sensor networks, internet of things, vehicular ad hoc networks and artificial intelligence.

Muhammad Aslam received his BS in Telecommunication System and his MS in Electrical Engineering from BZU Multan and COMSATS IIT Islamabad in 2010, 2012, respectively. He received his PhD from School of Computer Science of Dalian University of Technology, China. He is currently working as an Assistant Professor at COMSATS IIT, Wah Cantt. His major research interests are energy optimisation in WSNs, WBANs, UWSNs and SDWSNs.

Yao Nianmin is a Professor at School of Computer Science and Technology of Dalian University of Technology. He got his Bachelor, Master and Doctor degree from Jilin University. His main research interests include wireless ad hoc networks and artificial intelligence.

Naeem Ayoub is pursuing his PhD in Computer Science and Technology from Dalian University of Technology, China. He received his Masters in Information Technology from the University of Education Lahore, Multan, Pakistan. His research interests include computer vision, image processing, IoT and WSNs.

Khalid Ibrahim Qureshi is pursuing his PhD from School of Software Engineering, Dalian University of Technology, China. He received his Masters in Telecommunication and Networking from Bahria University Islamabad, Pakistan. His research interests include software defined networks, wireless communication and network security.

Ehsan Ullah Munir received his Masters in Computer Science from Barani Institute of Information Technology, Pakistan in 2001. He completed his PhD in Computer Software and Theory from Harbin Institute of technology Harbin, China in 2008. He is currently working as an Associate Professor in CIIT Wah, Pakistan. His research interests include heterogeneous parallel and distributed computing, wired and wireless networks, and information retrieval.

This paper is a revised and expanded version of a paper entitled 'HADCC: hybrid advanced distributed and centralized clustering path planning algorithm for WSNs' presented at *2014 IEEE 28th International Conference on Advanced Information Networking and Applications*, Victoria, BC, Canada, 13–16 May, 2014.

## 1 Introduction

Internet of things (IoT) provides the connectivity between physical devices such as, vehicles, building and all those things that contain actuators, sensors, embedded electronics or software (Zhang et al., 2014a). By enabling IoT, these intelligent items can collect and exchange information for multiple purpose more efficiently (Shen et al., 2017). IoT applications and concepts are emerging now a days which encourage researchers to put effort for industry and academia (Siow et al., 2018; Paul and Saraswathi, 2017; Lin et al., 2017). Interwork of wireless sensor networks (WSNs) and IoT brings reliability and efficient data flow between both networks (Chernyshev et al., 2017). Figure 1 shows the graphical view of inter-networking between WSNs and IoT.

Development of WSNs are based on micro electro mechanical systems (MEMS), system on chip (SoC), wireless communications and low power embedded systems (Zhang et al., 2015b, 2014b). Presently, WSNs are widely used in hostile region, military applications and security, remote territory, disaster areas and also in health sector (Bala et al., 2018). Particularly, WSNs consists of light-weight and small size sensor nodes that are used to supervise the environment or system. These sensor nodes become more advanced and intelligent when they are equipped with IoT enabled micro systems in them, so they can knob more complicated functions in WSNs technology (Baker et al., 2017). IoT enabled WSNs is one of the latest platform in today's smart energy management systems, it provides a dedicated gateway between sensor nodes and Base Station (BS) (Komalavalli, 2017). Once the sensor nodes are deployed in the network they cannot be recharged or replaced due to the high cost. Therefore, lifetime enhancement while maintaining the energy resources throughout the network operations are the essential for improving the performance of routing protocol in WSNs (Zhang and Zhang, 2012).

In order to manage the network, the concept of cluster and cluster-head (CH) is proposed which has an attractive methodology to make the network energy efficient (Haseeb et al., 2017; Zhang et al., 2017a; Dhiman and Singh, 2017; Awasthi et al., 2017; Ma et al., 2017). Classification of routing protocol is given in Figure 2. In clustering routing protocol, network is divided into clusters and each cluster is managed by CH. The selection of CH variate in

routing protocol but the major responsibilities of CH are the same in each routing protocol. CH is accountable for organising the cluster, management of routing table, collect data from all the associated member nodes, compressing and transmitting of data. Due to the more responsibilities, CHs consumes more energy than the computational energy of whole network (Ma et al., 2017; Zhang et al., 2015a, 2012).

**Figure 1** WSN assisted with IoT

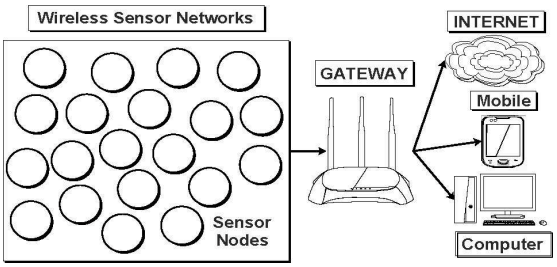
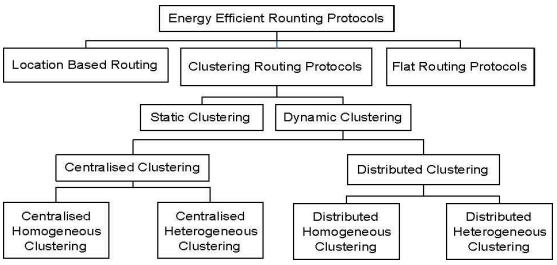



**Figure 2** Classification of energy efficient routing protocols




To summarise, managing the energy dissipation of CHs plays an important role in prolonging the network lifetime of routing protocol and archive energy efficiency.

Aiming towards higher energy efficient network, a new protocol named, multi adaptive clustering (MAC) is proposed to maintain the resources of sensor node in IoT enabled WSNs. The major contribution of this paper are as follows:

- We propose a new technique and architecture of MAC by enabling IoT which supports application specific requirement of WSNs.
- MAC provides better stability and better network lifetime due to the hybridisation of network. MAC also distribute the network load evenly among all the sensor nodes based on residual energy and distance to BS.
- Advance Cluster-head selection algorithm (CHSA) which can execute in a hybrid network and selects optimum number of CHs. This unique CHSA algorithm maintain centralised and distributed cluster formation which makes sure of higher packet delivery.
- MAC offers heterogeneity which makes it more suitable for large scale networks. Furthermore, proposed routing protocol based on realistic environment which helps it to perform more effectively in real-time scenarios.
- Experimental results show that MAC produce 204%, 119%, 56% and 72% higher stability period than LEACH-C, DEEC, D-DEEC and E-DEEC.

Particularly, the aforementioned contributions significantly extend the field of applications where the energy efficient routing protocol is required.

Remainder of this paper is organised as follows: In Section 2, the related work and motivations are briefly explained. System models and problem statement are defined in Section 3. Technical modelling of our proposed MAC are detailed in Section 4. Performance evaluation and evaluation metrics are given in Section 5. Simulation results are compared in Section 6. Results are analysed in Section 7. Characteristics comparison of proposed MAC with other protocols is given in Section 8. Conclusion and future works are drawn in Section 9.

## 2 Related work and motivations

In this section, we discuss the existing work on energy efficient schemes of WSNs, several works are proposed in the literature to extend the network lifetime of routing protocol in real-time scenarios of WSNs. We briefly describe the strength and weakness of some routing protocols in the following paragraph.

Low energy adaptive clustering hierarchy (LEACH) (Heinzelman et al., 2000) is the pioneer of hierarchical routing protocol in WSNs. The main contribution of LEACH is the dynamic selection of CHs using stochastic algorithm. following equation is used in stochastic algorithm to select a new CH in each round.

$$T_n = \begin{cases} \frac{P}{1-P(r \times mod \times \frac{1}{P})}, & \text{if } n \in N \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

This scheme of replacing CHs in each round divided the network load evenly on all nodes which results into significant amount of energy dissipation. Huge amount of data transmission leads to shortage of network lifetime because the nodes consumes more energy while transmission, this proofs the inefficiency of LEACH in large scale networks. Furthermore, LEACH is purely distributed network which lacks it the most of application specificity. Various version of LEACH are proposed in the past to improve the performance of LEACH.

LEACH-C (Centralised) (Heinzelman et al., 2002) is the well-known routing protocol, which is proposed to overcome the limitations of LEACH by using the central control algorithm. The network topology is same but the BS is in charge of generating the CHs by the knowledge of location and remaining energy of sensor nodes. BS calculates the average energy of network by this knowledge and compare the average energy with node's remaining energy. If a node has higher remaining energy than average energy then it will be selected as CH otherwise it will be a member node. In this way, energy dissipation in the selection process of CH and transmission overhead

**Table 1** Comparison of the state-of-the-art clustering routing protocols

| *Features of routing protocols* | *MAC* | *LEACH-C* | *DEEC* | *D-DEEC* | *E-DEEC* |
|---|---|---|---|---|---|
| Energy efficiency | ✓ | ✓ | | ✓ | |
| Heterogeneity-aware | ✓ | | | ✓ | ✓ |
| Scalability towards nodes | ✓ | | | ✓ | ✓ |
| Residual energy-based | ✓ | | ✓ | | ✓ |
| Application-aware | ✓ | ✓ | | ✓ | ✓ |
| Organised Network | ✓ | | ✓ | | |
| Evading rapid dissipation | ✓ | | | | ✓ |
| Avoiding redundant packets transmissions | ✓ | ✓ | ✓ | | ✓ |
| Controller assistance | ✓ | ✓ | ✓ | | |
| Complex algorithm | | ✓ | ✓ | ✓ | ✓ |
| Controlled CHs formation | ✓ | ✓ | ✓ | | |
| Hierarchal forwarding | ✓ | ✓ | | ✓ | |
| Higher data delivery | ✓ | | | | ✓ |
| Cooperative routing for long range transmission | ✓ | | ✓ | ✓ | |
| Inter-cluster communication | ✓ | | | | |
| Configuration through internet | ✓ | | | | |

are significantly reduced. However, the overhead still exist in large amount of data transmission furthermore, LEACH-C is not suitable for the network where the BS is located inside of network topology. Distributed energy efficient clustering (DEEC) (Qing et al., 2006) propose a distributed network with heterogeneous property by utilising two-level of heterogeneity. Similar like LEACH, DEEC also distributes the overall network load among all the sensor nodes by dividing the CH responsibilities. During the selection of CHs, DEEC does not need any global knowledge which is the key advantage of this protocol. The probability threshold of DEEC allows each node to take part in CH election.

$$T_{s_i} = \begin{cases} \frac{P_i}{1-P_i(r \times mod \times \frac{1}{P_i})}, \text{if } s_i \in G \ , \\ 0, \qquad \text{otherwise} \end{cases} \tag{2}$$

Developed-DEEC (D-DEEC) (Elbhiri et al., 2010) is proposed to balance the energy distribution among all the sensor nodes in the network. The method of CH selection is similar as DEEC in D-DEEC, the only difference in expression of selection CHs in advance nodes. The probability of being a CH is higher in advance nodes than normal nodes. Which results a drawback of routing protocol because the distributed algorithm always selects the advance nodes as a CH which leads to early death of these nodes and creates energy-hole in the network. Enhanced-DEEC (E-DEEC) (Saini and Sharma, 2010) proposed as an extension of DEEC routing protocol. E-DEEC propose three different types of nodes in the distributed network normal, advance and super nodes. Initial energy of these nodes increases in each type respectively, while the CH selection probability is in reverse order.

Each routing protocol proposed in the past tries to overcome the energy issue and improved some network lifetime (Zhang et al., 2016; Zhang, 2012; Zhang et al., 2013). Each routing protocol proposed different cluster formation with proper functional mathematical analysis (Zhang et al., 2017b; Misra et al., 2017; Fei et al., 2017; Carrabs et al., 2017; Kaur and Mahajan, 2018). Based on the above analysis, network performance can be enhanced with proper network topology, network coverage and knowledge of remaining energy and location of sensor nodes. In this paper we propose a new routing protocol; IoT Enabled Multi Adaptive Clustering energy efficient routing protocol supported by the new unique CHSA algorithm which selects optimum number of CHs in a hybrid network which brings better stability and improved network lifetime in simulations. Features comparison of proposed MAC routing protocols with state-of-the-art routing protocols is given in Table 1.

## 3 System models and problem statement

In this section, system models and network architecture are discussed in detail. Subsequently, the heterogeneous network and energy model are introduced. Moreover, problem definition and possible solution is described.

### *3.1 Problem statement*

As described in Section 1, due to the small size of sensor nodes WSNs has various limitations such as storage, computing, communicating and energy. As the requirement of longer network lifetime, most of the existing literature considered energy resources by minimising the energy dissipation through different techniques (Yetgin et al., 2017). By allowing WSNs to be an integral part of IoT, these sensor nodes can be controlled for all these resources. In this paper, we mainly focus on energy consumption and tried to extend the network lifetime with the help of IoT.

### *3.2 Heterogeneous network model*

MAC routing protocol considers two types of sensor nodes in order to sense the environment; normal nodes and super nodes. In the proposed model, normal nodes are referred

**Figure 3** Network design and architecture of multi adaptive clustering

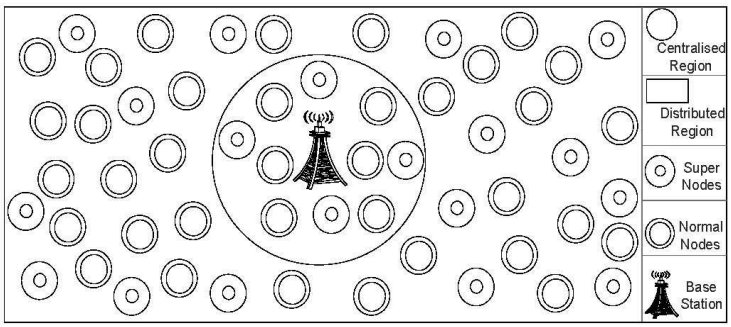
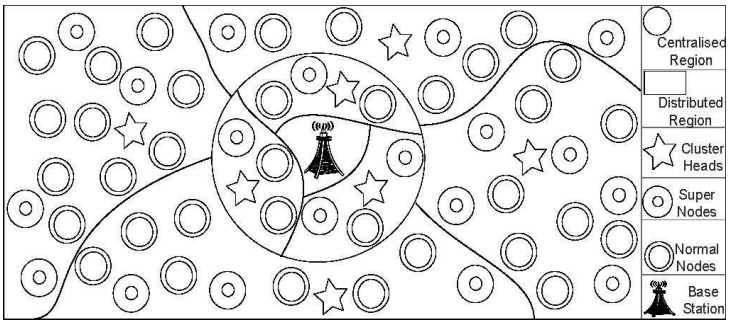



**Figure 4** Clustering mechanism and topology of multi adaptive clustering




as homogeneous nodes and super nodes are referred as heterogeneous nodes. The initial energy of normal nodes is set to 0.5J whilst the initial energy of advanced nodes is set to 1.0J. This extra property of heterogeneity brings more stability and robustness during selection of CHs because CHSA selects super nodes more often than normal nodes due to the higher remaining energy. The IoT brings advance scalability in MAC and allows to add more super nodes anytime during the network operations in order to increase the stability and network lifetime. To calculate the total initial energy of network, heterogeneous network model applies:

$$\begin{aligned} E_{IE} &= N_n(1-a)E_o + N_s N_n E_o(1+\alpha) \\ &= N_n E_o(1+\alpha N_s) \end{aligned} \tag{3}$$

In equation (3), $E_{IE}$ is the total initial energy of the network, $N_s$ represents the super nodes with initial energy $E_{o(1+\alpha)}$ and $(1-a)$ while $N_n$ represents the normal nodes with initial energy $E_o$.

### 3.3 Energy consumption model

Energy consumption model is used to measure the performance of MAC. Depending on the distance between transmitter and receiver, first order radio model and free space model are used for fair comparison with other routing protocols (Martinez-Sala et al., 2005; Yick et al., 2008). Parameters of radio models are $E_{\text{elec}} = 50$ nJ/bit, $E_{fs} = 10$ pJ/bit/m$^2 and E_{\text{amp}} = 0.0013$ pJ/bit/m$^4$. Thus, to send a $j-bit$ message to a distance $d$, the model applies:

$$E_{Tx}(j,d) = E_{Tx-elec}(j) + E_{Tx-amp}(j,d) \tag{4}$$

In equation (4), $E_{Tx-elec}$ represents the transmission energy while $E_{Tx-amp}$ represent the links on a long distance potentially affected by "multi-path" fading.

$$E_{Tx}(j,d) = \begin{cases} jE_{\text{elec}} + E_{fs}jd^2, & \text{if } d < d_{Th}. \\ jE_{\text{elec}} + E_{amp}jd^4, & \text{if } d \geq d_{Th}. \end{cases} \tag{5}$$

In equation (5), $E_{Tx}(j,d)$ is the total energy consumption for transmitting a $j-bit$ data packet from a source to destination $d$. The energy consumption in transmission is denoted by either $E_{fs}jd^2$ or $E_{\text{amp}}jd^4$ depending on the distance threshold $d_{Th}$ and the distance $d$ while $d \geq d_{Th}$ reflects the condition of 'free-space'.

In order to receive this data packet the model applies:

$$E_{Rx}(j) = E_{Rx-elec}(j) = jE_{\text{elec}} \tag{6}$$

In equation (6), $E_{Rx}(j)$ represents the total energy consumption for receiving a $j-bit$ data packet.

### 3.4 Network architecture and design

The network architecture of MAC is divided into two regions; first region is centralised and the second region is distributed. Normal and super nodes are deployed in both regions while the BS is located in the centre of the network topology represented in Figure 3. Whereas, Figure 4 represents clustering topology of MAC, both regions have their own clusters and CHs.

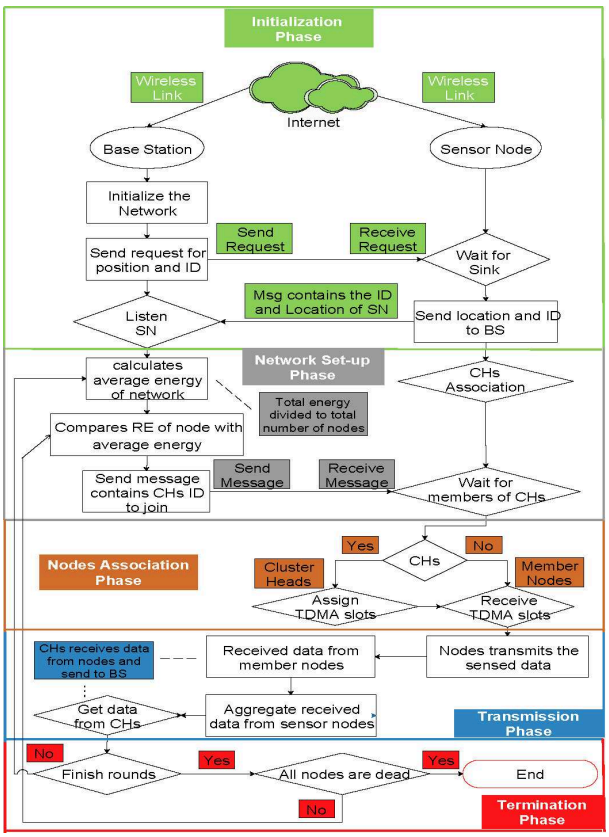
## 4 Proposal of IoT enabled multi adaptive clustering routing protocol

In this section, the MAC routing protocol is discussed in detail along with all major phases.

### 4.1 Proposed model

The proposed MAC features centralised and distributed clustering algorithms which brings the hybrid cluster formation. As we have discussed earlier, the network topology of MAC is divided into two regions and static sink is located in the centre of the network. MAC performs the cluster formation simultaneously in both regions; centralised and distributed. In order to execute and forming the clusters, MAC algorithm is divided into five major phases; network initialisation phase (NIP), network set-up phase (NSP), nodes association phase (NAP), data transmission phase (DTP) and network termination phase (NTP). All major phases of MAC along with communication flow is shown in Figure 5.

**Figure 5** Communication flow of multi adaptive clustering routing protocol (see online version for colours)




### 4.2 Network initialisation phase (NIP)

In the NIP of MAC, first the cluster formation is initialise in the centralised region where the BS controls the central algorithm while sensor nodes in the distributed region holds the control of algorithm and initialise the cluster formation. In the centralised region, a message containing node ID and location is sent to BS from each sensor node. The format of initial message is shown in Figure 6. The field $Message\ Type$ specifies that it is an initialisation message, $Source\ ID$ contains the node ID of transmitting node. The $X\text{-}coordinate$ and $Y\text{-}coordinate$ represents the abscissa and ordinate of node's location. The last field $Energy\ level$ indicates the remaining energy of sensor node.

**Figure 6** Format of initialisation message

| Message type | Source ID | X-Coordinate | Y-Coordinate | Energy Level |
|---|---|---|---|---|

Similarly, the same message is broadcasted in the distributed region and each node gets aware from other node's location and energy level.

### 4.3 Network set-up phase (NSP)

In NSP of MAC, BS calculates the distance between sensor node and BS itself and update the nodes table including node's location, and energy level. Then BS clusters the centralised region based on the distance and broadcast a $response\ message$ to the nodes in each cluster. Figure 7 shows the format of response message. The field $Message\ type$ indicates that this message is an association message for member nodes. The $Distance$ field is used to inform the maximum broadcasting range for all the sensor nodes in centralised region. The field $CH's\ ID$ is used to specify the ID of CH whilst the last field $Average\ Energy$ contains the network average energy.

**Figure 7** Format of response message

| Message type | Distance | CH's ID | Average Energy |
|---|---|---|---|

After receiving this response message, all sensor nodes in centralised region update this information in their routing table. After formation of clusters in the centralised region, the MAC starts cluster formation in the distributed region. When the nodes received the initialisation message, each node broadcast a message to become a CH, node with the highest energy level will be selected as CH and all the remaining sensor node will associate with the nearest CH and forms the clusters. Detailed CHs Selection Process (CHSP) for both regions is given below.

*CHSP in centralised region.* If there are $M$ CHs and each cluster have $N$ member nodes. Then each CH dissipates energy while receiving and transmitting the information. Therefore dissipated energy of each CH during communication will be calculated as:

$$M_{CH}(j,d) = jE_{\text{elec}}N_i + jE_{DA}(N_i) + 1 + jE_{\text{elec}} + E_{amp}jd_{toBS} \quad (7)$$

In equation (7), $jE_{\text{elec}}N_i$ represents the received signal from member nodes, $jE_{DA}(N_i)$ represents the data aggregation signal, $jE_{\text{elec}}$ shows the transmission of aggregated signal and $E_{amp}jd_{toBS}$ represents the distance from CH to BS. In order to localise member nodes BS broadcast initialisation message $IM$ to request member nodes for their location. Nodes response to BS and the message contains ID,

position, and the remaining energy of sensor node. By the knowledge of initial message, BS will calculate the energy consumption ratio (ECR) of each node by the following equation:

$$ECR_{N_i} = \frac{E_o}{E_o - E_r} \tag{8}$$

In equation (8), $E_o$ represents the initial energy and $E_r$ represents the residual energy of sensor node. After calculating the ECR, BS will able to calculate the suitability for final CH selection. In order to calculate the suitability, algorithm applies:

$$\text{Suitability} = E_r/(ECR \times dis_{to_{BS}}) \tag{9}$$

After calculating the suitability, BS will choose the perfect CHs by the desired percentage (10.0% in MAC) of being a CHs. Nodes with higher suitability will be selected as CHs.

*CHSP in distributed region.* After selection of CHs in centralised region, CHSA performs cluster formation in distributed region. When the formation executes in second region, the distributed property of CH selection will utilise and deploy all the nodes. The CHSA will adjust in the environment because the nodes are deployed away from the sink. Residual energy is an important metric for CH's selection in the second region. All nodes in the network broadcast the initialisation message in order to communicate with each other. Number of iteration and number of nodes to become CHs are symbolised as $i$ ,$M_i$ respectively and the CHSA rotates as an epoch.

When all the energy levels are calculated; than the final decision of CH will based upon threshold value. In order to calculate the threshold value, CHSA performs the following equation:

$$\text{Threshold} = \begin{cases} \frac{P_{M_i}}{(1-P_{M_i}(rand \times \frac{1}{P_{M_i}}) \times d)}, & \text{if } n \in \Upsilon \ , \\ 0, & \text{otherwise} \end{cases} \tag{10}$$

In equation (10), $\Upsilon$ represented as a set of nodes, $d$ is a distance between nodes and BS, and $r$ is current round. If $M_i$ is not selected as a CH in the running round than $n$ belongs to $\Upsilon$. In order to become a CH $M_i$ will choose a random number by itself from 0 to 1. CHSA will compare the random number with threshold, if the threshold is greater than the random number then $M_i$ will be chosen as a CH otherwise it will be a member node of that cluster. To calculate the energy level according to the desired percentage, the algorithm applies:

$$E_l = E_{\text{optimum}}/1 + \alpha \tag{11}$$

Energy level of super nodes which contains some extra energy than normal nodes will be calculated as:

$$P_{EE} = P_{\text{opt}}1 + \alpha/1 + \alpha M, \tag{12}$$

In the first centralised region the desired percentage will be calculated to become a CH is:

$$P_{CR} = P_{\text{opt}}N(1 + \alpha_i)/N + \sum_{i=1}^{N} \alpha i, \tag{13}$$

The above mention equations showing that the nodes having the slighter value of $epoch$ and have more energy with slighter distance to sink. These nodes have the higher chance to become a CH.

When CHs are chosen they will assign TDMA schedule to member nodes for communication.

$$\sum_{m=1}^{n} jE_{\text{elec}} + E_{amp}j(d_{tom}(m)) + jE_{\text{elec}} \tag{14}$$

BS assigns TDMA schedule to the CHs and when CHs receives data from member nodes they will transmit that data packet to the BS including their ID and remaining energy.

$$E_{mCH} = jE_{\text{elec}} + j_{ID}E_{\text{elec}} + jd_{tBS} \tag{15}$$

Algorithm 1 is proving that CH is taking the responsibility of load balancing which clearly leads it to increasing in the network lifetime.

**Algorithm 1** Cluster head selection algorithm (CHSA)

```
Initialization;
Set of nodes with a remaining energy in a cluster is
  recognised by ϒ;
Cluster head set ζ which is empty in initialization
  phase;
Radius of cluster is recognized by τ;
Energy consumption ratio is ECR = E_o/E_o − E_r;
Base station calculates the suitability;
while E_r/E_o/(E_o − E_r) × d do
    if suitability is greater than average energy then
        for Calculate Threshold do
            T = P_Mi/1 − P_Mi(rand × 1/P_Mi) × d;
            if threshold is greater than random number
              then
                tag is as a cluster head;
                set ζ = +1 ;
            member node;
        end
    member node;
end
```

### 4.4 Nodes association phase (NAP)

When the CHs are chosen for the current round than selected CHs will broadcast advertisement for node association using CSMA MAC (Medium Access Control) protocol. All CHs broadcast the association message in their cluster range and all the nodes in the network will stay active during the NAP, in this way sensor nodes will be able to choose the suitable CH. In order to associate with the CH, each node calculates the range by following equation:

$$Select_{criteria} = RSSI/Dis_{InRangeofCH}, \tag{16}$$

In equation (16), $RSSI$ represents the Received Signal Strength Indication and $Dis_{InRangeofCH}$ denotes the distance of CHs to the member nodes. When the nodes receive the advertisement from CHs they will transmit the

association request to their appropriate CH using CSMA medium access control (MAC) protocol. After receiving the association request CH will broadcast TDMA schedule to its corresponding nodes. When nodes get TDMA schedule they will consider it as a confirmations information that this node is associated to the particular cluster. CHs assigns TDMA slots to each node; each node will only transmit its sensed information on that time slot otherwise it will go the sleep mode.

### 4.5 Data transmission phase (DTP)

According to the MAC network topology the algorithm transmits data packets into time based iterations. So the node can only send its sensed data in the given time slots. Energy consumed by the node in one round can be calculated as:

$$E_{MN-CH} = jE_{\text{elec}} + jE_{fs}d^2_{toCH} \tag{17}$$

Nodes are randomly distributed in region $X \times Y$ and the occupied area by each cluster is $n^2/m$, in order to measure the distance from member nodes to CHs we use the following equation:

$$Dis_{toCH} = \int\int (x^2 + y^2) r(x, y) \mathrm{d}x\mathrm{d}y = \int\int \tau^2 r(\tau, \theta) \tau \mathrm{d}\tau \mathrm{d}\theta \tag{18}$$

After sending the data packet or when the node has no schedule of transmitting the data packet it will turn-off its transmitter and receiver; this is another property of saving energy. Until the working time of current running round; each CH will remain active all the time during the operations of network. Using multiple signal processing techniques CHs will compress the received data packets from their corresponding nodes and send the useful information to the BS. Energy consumed by the CH during the transmission of this message will be:

$$E_{CH} = jE_{\text{elec}}\left(\frac{n}{m} - 1\right) + jE_{DA}\frac{n}{m} + jE_{\text{elec}} + jE_{amp}d^2_{toBS} \tag{19}$$

In order to minimise the hot-spot area problem, we use super nodes act as relays for sensors which are located far from CHs. These nodes sensed their data and transmit it to super nodes. Super nodes will add this data with its own data and aggregate to the CH. To calculate the distance from the node to super node and CH, we use the simple local energy aware algorithm which is detailed in the following Algorithm 2.

### 4.6 Network termination phase (NTP)

After transmitting all the sensed data to BS, NTP checks the rounds status if the current running round is completed than NTP checks the alive nodes in the network otherwise algorithm go back to the NSP. If there is no alive node left in the network than the algorithm will terminate itself.

**Algorithm 2** Local energy aware algorithm

```
Initialization;
Set of member nodes with in a cluster is recognised by
 η;
while there is a message to be forwarded j do
    first calculate the distance from own location to
     cluster-head;
    if distance is greater than nearest node then
        check the table and compare energy level;
        find the nearest node which has higher energy
         level;
        tag that node as a Super node;
        transmit the sensed data to the super node;
    else
        send it to Cluster Head;
    end
end
```

## 5 Performance evaluation

In this section, we simulate our proposed model and compare it with state-of-the-art routing protocols namely; LEACH-C, DEEC, D-DEEC and E-DEEC.

### 5.1 Simulation parameters and methodology

We evaluate the performance of MAC routing protocol implemented with MATLAB v9.2. For fair comparison, we use the same probability of signal collision and interference in wireless channel as in existing routing protocols. Furthermore, we adopt the same Media Access Control (MAC) as in LEACH-C, DEEC, D-DEEC and E-DEEC.

**Table 2** Characterisation of simulation parameters

| *Parameter* | *Value* |
|---|---|
| Initial Energy of homogeneous nodes | .3J |
| Initial Energy of heterogeneous nodes | 1.0J |
| Percentage of being Cluster Head | 10.0% |
| $E_{DA}$ Energy cost | 50pJ/bitJ |
| Number of nodes | 100 |
| Number of rounds | 9999 |
| packet size | 4000 bit |
| $E_{TX}$ | 50nJ/bit |
| $E_{RX}$ | 50nJ/bit |
| $E_{fs}$ | 10pJ/bit/$m^2$ |
| $E_{amp}$ | 100pJ/bit/$m^2$ |

In order to explain the major simulation parameters, we dispersed 100 sensor nodes in $100 \times 100$ m network dimensions which are connected through internet with single static BS located in the centre of the network and considered 9999 number of rounds. Initial energy of homogenous sensor nodes is set to 0.5 J while heterogeneous sensor nodes are set to 1.0 J. Both homogeneous and heterogeneous sensor nodes are distributed randomly in centralised and distributed regions of network. Size of data packet per transmission is 4000 bits and the percentage to become CHs per round is

set to 10.0% in both regions. The detailed performance parameters that are used in our simulations are given in Table 2.

### 5.2 Evaluation metrics

The following performance metrics are considered for comparison:

- *Network lifetime or DLN (Death of Last Node)*: During the operation of network, sensor nodes utilises their energy for transmission and collection of data. These sensor nodes eventually starts dying after consuming all the energy. Network lifetime is measured in rounds and the current running round at the time of DLN will be considered as the maximum lifetime of network.
- *Network stability or DFN (Death of First Node)*: Network stability is directly proportional to networklifetime and it is the most important factor in energy efficient routing protocols. Current running round at the time of DFN is considered as the network stability, after the DFN network instability period starts.
- *Network instability Period (NIP)*: Total number of rounds/time from the DFN to DLN is the instability period of routing protocol.
- *Packets transmitted to CHs (Cluster Heads)*: After collecting the data from sensing field, sensor nodes transmit this data towards CHs. Maximum received data packets by CHs strengthen the routing protocol.
- *Packets transmitted to BS (Base Station)*: After receiving the data from sensor nodes, CHs compress all the data and forward it to BS. Maximum received data packets by BS shows the robustness of routing protocol.
- *CHs per round*: Minimum generation of CHs in single round while maintaining the requirement of network helps in prolonging the network lifetime.
- *Packet delivery ratio (PDR)*: Communication ability of routing protocols can be measured by calculating the PDR. Ratio of the successfully received data packets from the total number of transmitted data packets is the PDR of any routing protocol.

## 6 Comparison results

In this section, we compare the MAC routing protocol with other four protocols (LEACH-C, DEEC, D-DEEC and E-DEEC). All five protocols are deeply analysed with aforementioned evaluation metrics. Figures 6–10 shows the comparison of proposed protocol with above mentioned other protocols. The detailed comparison of all the five protocols is as follows.

### 6.1 Number of alive nodes

Number of alive nodes is the most important measurement metric for any routing protocol in WSNs. Figure 8. shows the evidence of improvement in proposed MAC protocol in the comparison of existing routing protocols. The network lifetime of proposed MAC protocol is improved by 7124 rounds, 5600 rounds, 3339 rounds and 5013 rounds from LEACH-C, DEEC, D-DEEC and E-DEEC respectively. Nodes in LEACH-C die very early because of large amount of communication control messages, whilst DEEC and E-DEEC perform quite better than LEACH-C but due to the lack of resources and long range communication these protocols do not survive for longer period. Whereas, the sensor nodes in D-DEEC survived till the 5128 rounds which is best among other routing protocols but the network using MAC has a very long network lifetime which performs efficiently till 8467 rounds.

**Figure 8** Number of alive nodes (see online version for colours)

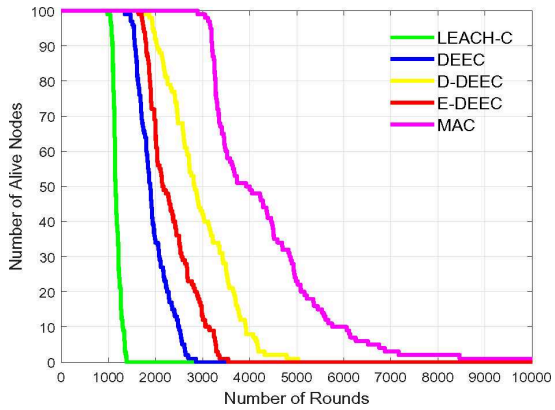
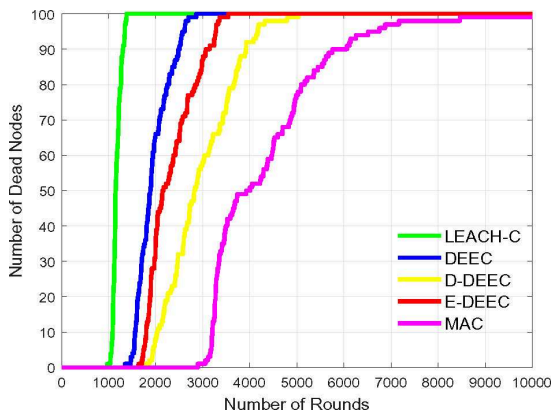



**Figure 9** Number of dead nodes (see online version for colours)




### 6.2 Number of dead nodes

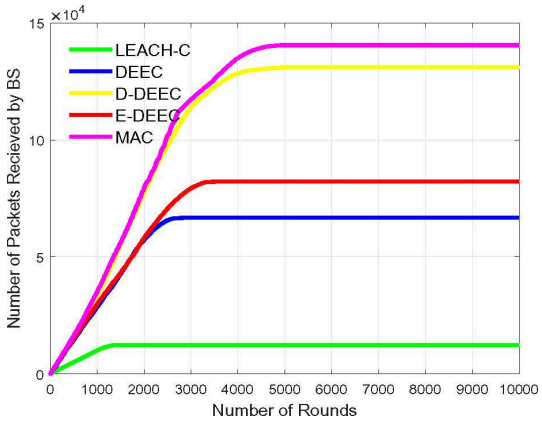
Number of dead nodes is used to measure the stability period of any routing protocol in WSNs. Figure 9 shows the number of dead nodes with respect to number of rounds. The difference of network stability between MAC are 1995

rounds, 1618 rounds, 1075 rounds and 1245 rounds in the comparison of LEACH-C, DEEC, D-DEEC and E-DEEC respectively. The DFN in proposed protocol is at 2972 round which is 204%, 119%, 56% and 72% better than other protocols respectively.

### 6.3 *Number of packets transmitted towards base-station (BS)*

Figure 10 shows the number of total data packets transmitted and received by the BS. Figure shows that D-DEEC delivers considerable data packets in the comparison of LEACH-C, DEEC and E-DEEC. But the data packets received in the network of MAC is higher than D-DEEC. This is due to the local energy aware algorithm in the proposed model, when the distance between CH is greater than the super nodes then the super node will act as a CH which helps to deliver the maximum data packets without delay. However, after the 2972 rounds, MAC packet delivery decreases due to the settings of local energy aware algorithm. When more nodes starts dying in the network, the distance between sensor nodes will increase then the packets received by BS faces considerable amount of delay.

**Figure 10** Packets transmission to BS (see online version for colours)




### 6.4 *Number of packets transmitted towards cluster-heads*

Figure 11 shows that the packet delivered to CHs by MAC protocol is better than other protocols; MAC protocol consumes less energy which results into maximising in the lifespan of network and also increasing in the packet delivery towards CHs.

### 6.5 *Number of cluster-heads generated*

Figure 12 vividly shows the number of CHs generated with respect to number of rounds. When the percentage of being a CH in network setting is small then the sensor nodes begin to die very early while when the percentage of CHs is large then the whole network performs poorly such as, sudden energy dissipation and smaller data transmission which results into shorter lifespan of network. Taking all these aspects into consideration, percentage of being a CHs in proposed MAC protocol is set to 10.0% which generates perfect CHs to lead the network. This percentage of CHs covers the whole network area and distribute the energy resources equally among all the sensor nodes.

**Figure 11** Packets transmission to CHs (see online version for colours)

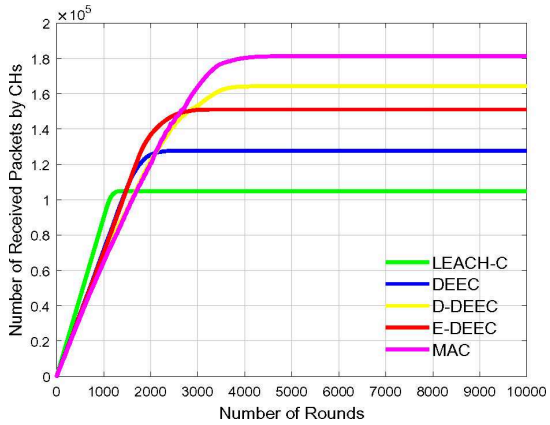
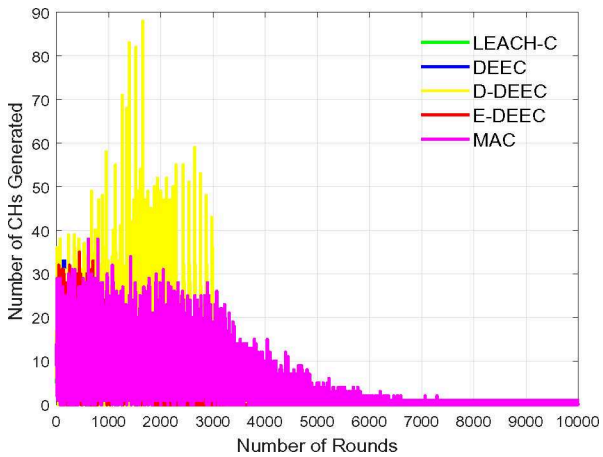



**Figure 12** CHs generated per round (see online version for colours)




## 7 Results analysis in different size of networks

In this section, we analyse the proposed model in the comparison of LEACH-C, DEEC, D-DEEC and E-DEEC. To improve the quality of graphs, first we simulate all the routing protocols in MATLAB v9.2 and then generate the graphs in OriginLab for deep analysis.

The simulation in this section, consists of 100 sensor nodes which are dispersed in $100 \times 100$ m, $200 \times 200$ m and $300 \times 300$ m, respectively. The rest of the simulation parameters are same as mentioned in Table 2. The reason of this simulation is to prove the performance of proposed protocols in large scale networks.

**Table 3** Characteristic comparison of proposed protocol with state-of-the-art routing protocols

| *Routing protocols* | *LEACH-C* | *DEEC* | *D-DEEC* | *E-DEEC* | *MAC* |
|---|---|---|---|---|---|
| Death of first node (DFN) | 977 rounds | 1354 rounds | 1897 rounds | 1727 rounds | 2972 rounds |
| Death of last node (DLN) | 1343 rounds | 2867 rounds | 5128 rounds | 3454 rounds | upto 8467 rounds |
| NIP | 366 rounds | 1513 rounds | 3231 rounds | 1727 rounds | upto 5495 rounds |
| Packet Delivery Ratio | Bad | Better | Better | Good | Best |
| Scalability | Bad | Bad | Good | Better | Best |
| Localisation-Awareness | Yes | Yes | No | No | Yes |
| Communication Path | Single-Hop | Single-Hop | Single-Hop | Single-Hop | Single-Hop |
| BS Location | Fixed at centre | Fixed at center | Fixed at centre | Fixed at centre | Fixed at centre |

Figure 13 vividly shows the network lifetime of proposed protocol in different size of networks. Due to the small amount of sensor nodes, network lifetime of all five protocols decreases gradually in larger networks. Similarly, Figure 14 shows the network stability period of all five routing protocols. It can be seen clearly that proposed MAC routing protocol perform much better in comparison of LEACH-C, DEEC, D-DEEC and E-DEEC. Figures proved that MAC routing protocol is suitable for large scale of networks in terms of network lifetime and network stability.

**Figure 13** Network lifetime (see online version for colours)

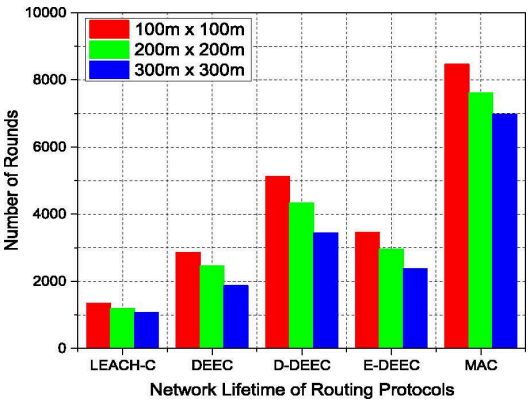
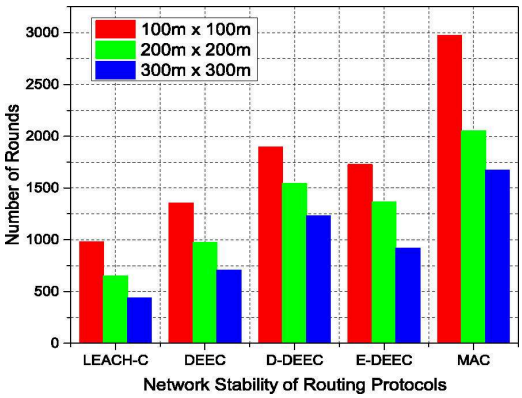



**Figure 14** Stability of network (see online version for colours)

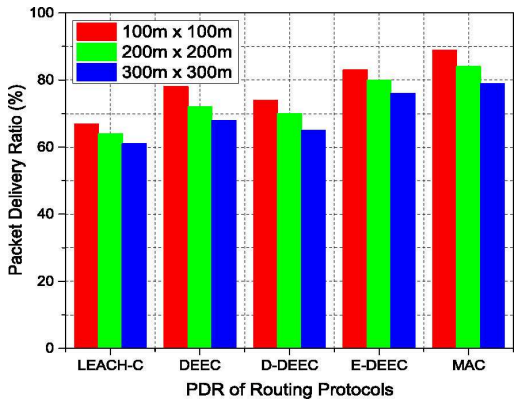



During the transmission of data packets towards BS, it is assumed that all data packets are successfully received by the BS but in reality, due to the noise, reflection, interference and attenuation some data packets are dropped. In order to perform the fair comparison, we use the uniform random distribution model to calculate the packet delivery ratio (Zhou et al., 2009). The probability of packet delivery fluctuate due to the change of distance between CHs and BS because the CHs are changing with respect to rounds. To measure the accurate packet delivery ratio we use the following equation:

$$PDR = \begin{cases} 1 & \text{if } D_{CH,BS} > 100 \\ 1/20 \times (D_{CH,BS} - 80) & \text{if } 80 \leq D_{CH,BS} \leq 100 \\ 0 & \text{if } 0 \leq D_{CH,BS} < 80 \end{cases} \tag{20}$$

where $D_{CH,BS}$ shows the distance between CH and BS, if the probability is larger than PDR, than the packet will successfully delivered otherwise it will be dropped. Figure 15 shows the number of packets which are delivered successfully. Figure proves that proposed MAC routing protocol has the highest packet delivery ratio not only in standard network dimensions but also in larger network.

**Figure 15** Packet delivery ratio (see online version for colours)




## 8 Characteristics comparison of MAC and existing conventional routing protocols

The characteristics comparison over performance of LEACH-C, DEEC, D-DEEC, E-DEEC and MAC are presented in Table 3. The characteristics used to compare the performance are DLN, DFN, NIP, PDR, scalability, localisation-awareness, communication path and BS location. As shown in Table 3, MAC performs better than LEACH-C, DEEC, D-DEEC and E-DEEC

in terms of DLN, DFN, NIP, PDR, scalability, and localisation-awareness.

However, for the case of communication path and BS location, MAC routing protocol is suitable for the network which contains the single-hop communication and where the BS is static and located in the centre of the network.

## 9 Conclusion and future work

In this paper, we propose a new technique of WSNs namely; multi adaptive clustering (MAC) that is assisted with IOT to enable the resources in sensor nodes. Proposed routing technique consists of a new hybrid network architecture which contains centralised and distributed network where the homogeneous and heterogenous nodes are deployed and the BS is located in the centre of hybrid network. MAC also propose an advanced CH selection algorithm (CHSA) which efficiently generates optimum number of CHs in both network by calculating the residual energy of sensor nodes. We compare the performance of MAC with state-of-the-art routing protocols such as; LEACH-C, DEEC, D-DEEC and E-DEEC. Simulation results and performance evaluation of MAC provide the evidence of significant improvement in terms of network lifetime, network stability and packet delivery ratio. In our future work, we would like to implement MAC in NS-3 simulation environment which will give the more accurate picture of performance improvement. The real-time implementation of MAC can also be considered for future work.

## References


Awasthi, K.K., Singh, A.K. and Tahilyani, S. (2017) 'Study of cluster based routing protocols in wireless sensor networks', *Optimization*, Vol. 9, p.10.

Baker, T., Asim, M., Tawfik, H., Aldawsari, B. and Buyya, R. (2017) 'An energy-aware service composition algorithm for multiple cloud-based IoT applications', *Journal of Network and Computer Applications*, Vol. 89, pp.96–108.

Bala, T., Bhatia, V., Kumawat, S. and Jaglan, V. (2018) 'A survey: issues and challenges in wireless sensor network', *International Journal of Engineering and Technology (UAE)*, Vol. 7, No. 2.4, pp.53–55.

Carrabs, F., Cerulli, R., D'Ambrosio, C. and Raiconi, A. (2017) 'Prolonging lifetime in wireless sensor networks with interference constraints', *International Conference on Green, Pervasive, and Cloud Computing*, May, Springer, Cham, pp.285–297.

Chernyshev, M., Baig, Z., Bello, O. and Zeadally, S. (2017) 'Internet of things (IoT): research, simulators, and testbeds', *IEEE Internet of Things Journal*, Vol. 5, No. 3.

Dhiman, S. and Singh, O. (2017) 'Energy efficient routing protocol for wireless sensor networks: a review', *IJASRE*, p.3.

Elbhiri, B., Saadane, R. and Aboutajdine, D. (2010) 'Developed dISTRIBUTED energy-efficient clustering (DDEEC) for heterogeneous wireless sensor networks', *5th International Symposium on I/V Communications and Mobile Network (ISVC, 2010)*, September, IEEE, pp.1–4.

Fei, Z., Li, B., Yang, S., Xing, C., Chen, H. and Hanzo, L. (2017) 'A survey of multi-objective optimization in wireless sensor networks: metrics, algorithms, and open problems', *IEEE Communications Surveys and Tutorials*, Vol. 19, No. 1, pp.550–586.

Haseeb, K., Bakar, K.A., Abdullah, A.H. and Darwish, T. (2017) 'Adaptive energy aware cluster-based routing protocol for wireless sensor networks', *Wireless Networks*, Vol. 23, No. 6, pp.1953–1966.

Heinzelman, W. R., Chandrakasan, A. and Balakrishnan, H. (2000) 'Energy-efficient communication protocol for wireless microsensor networks', *Proceedings of the 33rd Annual Hawaii International Conference on System Sciences, 2000*, January, 10-pp, IEEE.

Heinzelman, W.B., Chandrakasan, A.P. and Balakrishnan, H. (2002) 'An application-specific protocol architecture for wireless microsensor networks', *IEEE Transactions on Wireless Communications*, Vol. 1, No. 4, pp.660–670.

Kaur, S. and Mahajan, R. (2018) 'Hybrid meta-heuristic optimization based energy efficient protocol for wireless sensor networks', *Egyptian Informatics Journal*.

Komalavalli, C. (2017) 'Convergence of wireless sensor networks, internet of things, big data: challenges', *Convergence*, Vol. 6, No. 6.

Lin, J., Yu, W., Zhang, N., Yang, X., Zhang, H. and Zhao, W. (2017) 'A survey on internet of things: architecture, enabling technologies, security and privacy, and applications', *IEEE Internet of Things Journal*, Vol. 4, No. 5, pp.1125–1142.

Ma, Z., Zhang, D.G., Chen, J. and Hou, Y.X. (2017) 'Shadow detection of moving objects based on multisource information in internet of things', *Journal of Experimental and Theoretical Artificial Intelligence*, Vol. 29, No. 3, pp.649–661.

Martinez-Sala, A., Molina-Garcia-Pardo, J.M., Egea-Ldpez, E., Vales-Alonso, J., Juan-Llacer, L. and Garcia-Haro, J. (2005) 'An accurate radio channel model for wireless sensor networks simulation', *Journal of Communications and Networks*, Vol. 7, No. 4, pp.401–407.

Misra, S., Chatterjee, S. and Obaidat, M.S. (2017) 'On theoretical modeling of sensor cloud: A paradigm shift from wireless sensor network', *IEEE Systems Journal*, Vol. 11, No. 2, pp.1084–1093.

Paul, P.V. and Saraswathi, R. (2017) 'The internet of things: a comprehensive survey', *International Conference on Computation of Power, Energy Information and Commuincation (ICCPEIC, 2017)*, March, IEEE, pp.421–426.

Qing, L., Zhu, Q. and Wang, M. (2006) 'Design of a distributed energy-efficient clustering algorithm for heterogeneous wireless sensor networks', *Computer Communications*, Vol. 29, No. 12, pp.2230–2237.

Saini, P. and Sharma, A. K. (2010, October) 'E-DEEC-enhanced distributed energy efficient clustering scheme for heterogeneous WSN', *1st International Conference on Parallel Distributed and Grid Computing (PDGC, 2010)*, IEEE, pp.205–210.

Shen, J., Wang, A., Wang, C., Hung, P.C. and Lai, C.F. (2017) 'An efficient centroid-based routing protocol for energy management in WSN-assisted IoT', *IEEE Access*, Vol. 5, pp.18469–18479.

Siow, E., Tiropanis, T. and Hall, W. (2018) 'Analytics for the internet of things: a survey', *ACM Computing Surveys (CSUR)*, Vol. 51, No. 4, p.74.

Yetgin, H., Cheung, K.T.K., El-Hajjar, M. and Hanzo, L.H. (2017) 'A survey of network lifetime maximization techniques in wireless sensor networks', *IEEE Communications Surveys and Tutorials*, Vol. 19, No. 2, pp.828–854.

Yick, J., Mukherjee, B. and Ghosal, D. (2008) 'Wireless sensor network survey', *Computer Networks*, Vol. 52, No. 12, pp.2292–2330.

Zhang, D-g. and Zhang, X-d. (2012) 'Design and implementation of embedded un-interruptible power supply system EUPSS for web-based mobile application', *Enterp. Inf. Syst.*, Vol. 6, No. 4, November, pp.473–489.

Zhang, D., Kang, X. and Wang, J. (2012) 'A novel image de-noising method based on spherical coordinates system', *EURASIP Journal on Advances in Signal Processing*, Vol. 2012, No. 1, p.110.

Zhang, D.G. and Liang, Y.P. (2013) 'A kind of novel method of service-aware computing for uncertain mobile applications', *Mathematical and Computer Modelling*, Vol. 57, Nos. 3–4, pp.344–356.

Zhang, D., Wang, X., Song, X. and Zhao, D. (2014a) 'A novel approach to mapped correlation of ID for RFID anti-collision', *IEEE Transactions on Services Computing*, Vol. 7, No. 4, pp.741–748.

Zhang, D., Li, G., Zheng, K., Ming, X. and Pan, Z.H. (2014b) 'An energy-balanced routing method based on forward-aware factor for wireless sensor networks', *IEEE Transactions on Industrial Informatics*, Vol. 10, No. 1, pp.766–773.

Zhang, D.G., Wang, X., Song, X.D., Zhang, T. and Zhu, Y.N. (2015) 'A new clustering routing method based on PECE for WSN', *EURASIP Journal on Wireless Communications and Networking*, Vol. 2015, No. 1, p.162.

Zhang, D-G., Zheng, K., Zhang, T. and Wang, X. (2015b) 'A novel multicast routing method with minimum transmission for WSN of cloud computing service', *Soft Comput.*, Vol. 19, No. 7, July, pp.1817–1827.

Zhang, D. G., Zheng, K., Zhao, D.X., Song, X.D. and Wang, X. (2016) 'Novel quick start (QS) method for optimization of TCP', *Wireless Networks*, Vol. 22, No. 1, pp.211–222.

Zhang, D.G., Liu, S., Zhang, T. and Liang, Z. (2017a) 'Novel unequal clustering routing protocol considering energy balancing based on network partition & distance for mobile education', *Journal of Network and Computer Applications*, Vol. 88, pp.1–9.

Zhang, J., Tang, J., Wang, T. and Chen, F. (2017b) 'Energy-efficient data-gathering rendezvous algorithms with mobile sinks for wireless sensor networks', *International Journal of Sensor Networks*, Vol. 23, No. 4, pp.248–257.

Zhang, D.G. (2012) 'A new approach and system for attentive mobile learning based on seamless migration', *Applied Intelligence*, Vol. 36, No. 1, pp.75–89.

Zhou, Q., Cao, X., Chen, S. and Lin, G. (2009) 'A solution to error and loss in wireless network transfer', *International Conference on Wireless Networks and Information Systems, 2009. WNIS'09*, IEEE, December, pp.312–315.